

[Dashboard](#) / [My courses](#) / [MA-224-G 24H](#) / [Tests](#)/ [Test 4 \(topics 10-12: Applications of Lagrange's theorem, Coding theory, Coding theory and group codes\)](#)**Started on** Wednesday, 27 November 2024, 2:17 PM**State** Finished**Completed on** Wednesday, 27 November 2024, 2:37 PM**Time taken** 20 mins 13 secs**Marks** 0.83/3.00**Grade** 0.83 out of 3.00 (27.8%)

Information

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This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Incorrect

Mark 0.00 out of 1.00

If p and q be prime numbers, let $n = p \cdot q$ and write (n, x) and (n, y) for the public and private key in an RSA key pair associated to n .

In the list below we have written suggestions for $(x, y, p, q, p \cdot q)$. Some of them give valid RSA keys, some of them do not for different reasons.

Select all options that give us a valid key pair. Leave the invalid options un-checked.

- ☐ $x = 3008046, y = 2067287, p = 3881, q = 2549, n = 9892669$
- ☒ $x = 2497423, y = 16962307, p = 7231, q = 2687, n = 19429697$
- ☒ $x = 20271383, y = 22663367, p = 3557, q = 7741, n = 27534737$
- ☒ $x = 26924209, y = 43485577, p = 7239, q = 7789, n = 56384571$
- ☐ $x = 27087494, y = 8924683, p = 7331, q = 4231, n = 31017461$
- ☒ $x = 35519203, y = 9772867, p = 5471, q = 7351, n = 40217321$
- ☐ None of the above

You are free to use this calculator for products $\text{mod } (n)$. For example, if you want to compute $(7634 \cdot 99833 \% 8883993)$ you should set **Factor 1**=7634, **Factor 2**=99832, and **Modulus**=8883993.

Factor 1:
factor 2:
Modulus:
Result:

You are free to use this calculator for computing powers $\text{mod } (n)$. For example, if you want to compute $(7634^{99383} \% 8883993)$, you should set **Base**=7634, **Exponent**=99382, and **Modulus**=8883993.

Base:
Exponent:
Modulus:
Result:

Question 2

Partially correct

Mark 0.50 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

[110010, 011110, 001001, 000000]

.

$l_{min} =$

Your last answer was interpreted as follows:

2

$l_{max} =$

Your last answer was interpreted as follows:

4

Question 3

Partially correct

Mark 0.33 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be the encoding function given by $E(w_1w_2w_3) = w_1w_2w_3w_4w_5$, where

$$\begin{bmatrix} w_4 = w_3 + w_1 \\ w_5 = w_3 + w_2 + w_1 \end{bmatrix}$$

a) Find the generator matrix G for the encoding E

 $G =$

1	0	0	1	1
0	1	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

b) Find the parity check matrix H for the encoding E .

 $H =$

1	0	0	1	0
0	1	0	0	0

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

c) Let $r = 01011$ be a received message. Compute the syndrome of r with respect to the encoding E .

Syndrome of $r =$ 1111

Your last answer was interpreted as follows:

1111

Syntax guide: Give your answers to **a)** and **b)** as matrices. Use a space character to separate entries in the same row, and line break to separate rows.

Syntax guide: Give your answer to **c)** as a bitstring. E.g. if you think the answer to a question is "1011", then you should enter **1011**.

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 541

Signing code:

Your last answer was interpreted as follows:

17885

[◀ Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)

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